



Faculty of Engineering & Technology
Computer Science Department
Software Engineering
COMP433

Phase Three: Requirement Analysis & Modelling

Business: Vet Doctor – Veterinary Home Visit Service

Prepared by: Developer Group 10

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Section: 3

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Task 3.1: Actor analysis

Lead: | Contributors: (reviewing), (discussion), (diagram), (finalising)

ACTOR	SEMANTICS / DESCRIPTION
GUEST	This actor represents an unregistered visitor who can access public informational content including service descriptions, booking guides, and frequently asked questions, but cannot book appointments or access any personalised services
CLIENT	This actor represents a registered user who seeks veterinary services, and can register an account, log in, update account information, book veterinary visits, make payments, track visit status, communicate with the assigned veterinarian, and submit ratings and reviews following completed visits
VETERINARIAN	This actor represents a registered and administrator-approved professional who can manage their availability calendar, conduct visits, update appointment status, communicate with clients, submit post-visit medical records and prescriptions, and can only view medical records of animals linked to their own appointments
ADMINISTRATOR	This actor represents the system owner who can approve or reject veterinarian registrations, manage and update user accounts, configure service pricing and policies, moderate reviews, and monitor system performance. The administrator account is pre-configured and cannot be registered publicly
PAYMENT GATEWAY	This actor represents an external third-party system that processes payment transactions on behalf of clients and returns approval or rejection results to the system. It does not store or access any client data beyond the payment transaction itself
NOTIFICATION SERVICE	This actor represents an external system that delivers automated notifications to clients and veterinarians including booking confirmations, appointment reminders, and password reset links. It does not initiate any action independently and only responds to triggers from the system

Task 3.2 : Use-Case Modelling

Lead: | Contributors: (reviewing), (discussion), (diagram), (finalising)

Identified Use Cases

Use Case Diagram

Task 3.3

Task 3.3: Use Case Specifications

1 Use Case: Book Veterinary

Individual Task —

System Name : Vet Doctor – Veterinary Home Visit Service	
Use Case Title	BOOK A VISIT
Brief Description	A Client may book a veterinary home or farm visit by specifying the service type, animal details, preferred date and time, and visit location. The system assigns an available qualified veterinarian automatically or based on the client's selection. A booking confirmation is sent to both the client and the assigned veterinarian upon successful booking.
Actors	Client, Notification Service
Data	Service type, animal type, reason for visit, visit location, preferred date and time, veterinarian preference
Stimulus / Trigger	User command issued by the Client upon selecting "Book a Visit" from the dashboard
Pre-conditions	1. The Client is registered and logged into the system 2. At least one qualified veterinarian is available for the selected service type and date
Flow of Events	1. The Client selects "Book a Visit" from the dashboard 2. The system displays the available service types: Routine Check-Up, Emergency Visit, Farm Visit 3. The Client selects a service type and enters animal details and reason for the visit 4. The system displays a calendar showing only currently available date and time slots 5. The Client selects a preferred date and time and confirms the visit location 6. The Client either selects a specific veterinarian from a filtered list or opts for automatic assignment 7. If automatic assignment: the system assigns the nearest available qualified veterinarian based on the client's location and service type 8. If the preferred slot is unavailable: the system suggests the next 3 available slots for the client to choose from 9. The system presents a booking summary for the client to review 10. The Client confirms the booking 11. The system creates the appointment record and triggers a confirmation notification to both the Client and the assigned Veterinarian If Emergency Visit: the system flags the booking as high priority and requires the Veterinarian to acknowledge within 15 minutes
Post-conditions / Response	1. A new appointment record is created and stored in the system 2. The assigned Veterinarian receives a booking notification

	3. The Client receives a booking confirmation including date, time, location, and service type
Comments	Emergency Visit bookings are flagged as high priority. Cancellations made less than 2 hours before the scheduled visit trigger a cancellation fee notification to the Client as configured by the Administrator

2 Use Case: Login

Individual Task —

System Name : Vet Doctor – Veterinary Home Visit Service	
Use Case Title	LOG IN
Brief Description	A registered Client, Veterinarian, or Administrator may log into the system by providing their registered email address and password. The system validates the credentials and grants the actor access to their role-specific dashboard. If credentials are invalid, the system rejects the login attempt and notifies the actor.
Actors	Client, Veterinarian, Administrator, Notification Service
Data	Registered email address, password
Stimulus / Trigger	User command issued by the actor upon submitting the login form
Pre-conditions	1. The actor has a registered account in the system 2. The account is active and not suspended or locked
Workflow / Sequence of Events	1. The actor navigates to the login page 2. The actor enters their registered email address and password 3 The system validates the provided credentials against the database 4 If valid: the system authenticates the actor, creates a session, and redirects to the role-specific dashboard 5 If invalid: the system displays an error message and prompts the actor to re-enter credentials If 5 consecutive failed attempts occur: the system locks the account for 15 minutes and the Notification Service sends a lockout email to the registered address If the actor selects "Forgot Password": the system sends a secure time-limited reset link (valid for 30 minutes) to the registered email via the Notification Service If the reset link expires: the actor must submit a new password reset request
Post-conditions / Response	1. The actor is authenticated and granted access to their role-specific dashboard 2. A session is created for the authenticated actor 3. Failed login attempts are logged by the system

Comments	Veterinarian accounts pending Administrator approval cannot log in until the account is activated. Passwords are stored using bcrypt hashing and are never transmitted in plain text
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3 Use Case: Make Payment

Individual Task —

System Name : Vet Doctor – Veterinary Home Visit Service	
Use Case Title	MAKE PAYMENT
Brief Description	A registered Client may pay for veterinary services using one of the supported payment methods: Cash on Delivery, Credit/Debit Card via a PCI-DSS compliant gateway, or a supported E-wallet. The system calculates the total invoice, processes the payment, generates a digital invoice, and updates the appointment record accordingly.
Actors	Client, Payment Gateway, Notification Service
Data	Payment method, card or wallet details, service charges, additional charges entered by the Veterinarian, invoice details
Stimulus / Trigger	User command issued by the Client upon selecting a payment method at the time of booking or upon service completion
Pre-conditions	1. The Client is registered and logged into the system 2. The Client has a confirmed or completed appointment associated with an outstanding payment
Workflow / Sequence of Delivery, Events	1. The system presents the Client with available payment options: Cash on Credit/Debit Card, or E-wallet 2. The Client selects the preferred payment method 3. If Credit/Debit Card or E-wallet: the system redirects to the PCI-DSS compliant Payment Gateway 4. The Client enters the required payment details 5. The Payment Gateway processes the transaction and returns an approval or rejection result to the system 6. The system calculates the total invoice based on the base service price plus any additional charges entered by the Veterinarian 7. If additional charges exist: the system prompts the Client to acknowledge them before generating the final invoice 8. Upon successful payment: the system generates a digital itemized invoice and sends it to the Client's registered email via the Notification Service 9. The invoice is also displayed within the Client's account If Cash on Delivery: the invoice is generated after the Veterinarian marks the visit as Completed and confirms cash receipt
Post-conditions / Response	1. The payment is recorded and associated with the appointment 2. A digital invoice is generated and sent to the Client's registered email 3. The invoice is stored securely and accessible from the Client's transaction history

Comments	The system never stores, logs, or transmits card details in plain text. Only a masked reference (last 4 digits) is retained for display purposes. Clients may filter and download past invoices as PDF files from their dashboard
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4 Use Case: Manage Appointment Information

Individual Task —

System Name : Vet Doctor – Veterinary Home Visit Service	
Use Case Title	MANAGE APPOINTMENT-RELATED INFORMATION
Brief Description	A registered and approved Veterinarian may manage all appointment-related information including their schedule, visit statuses, post-visit medical records, and prescriptions. The system allows the Veterinarian to update appointment status through defined stages, submit post-visit records, flag appointments for follow-up, manage their availability calendar, and view the full medical history of animals linked to their appointments.
Actors	Veterinarian, Notification Service
Data	Appointment details, visit status, post-visit notes, digital prescription, uploaded files (photos, lab results), follow-up date, availability calendar slots, animal medical history
Stimulus / Trigger	User command issued by the Veterinarian upon navigating to the appointment management section of their dashboard
Pre-conditions	1. The Veterinarian is registered, approved by the Administrator, and logged into the system 2. The Veterinarian has at least one assigned appointment in the system

- Workflow / Sequence of upcoming Events
1. The Veterinarian navigates to their personalized dashboard displaying appointments sorted chronologically
 2. The system displays appointments for the day, week, or month with the service type shown as a label on each entry
 3. The Veterinarian selects an appointment and updates its status through the defined stages: Confirmed, En Route, In Progress, or Completed
 4. The system automatically notifies the Client via the platform whenever the appointment status changes
 5. Upon marking a visit as Completed: the system prompts the Veterinarian to upload visit notes (free text, max 2,000 characters), a digital prescription (PDF or structured form), and any relevant files (photos, lab results; max 10 MB per file)
 6. The uploaded records are securely stored and attached to the client's animal profile and made accessible to the Client through their account
 7. If the Veterinarian wishes to flag the appointment for follow-up: they specify a follow-up date and the system automatically sends a reminder notification to both the Veterinarian and the Client on that date
 8. The Veterinarian may also set or update their availability calendar by marking specific days or time slots as unavailable
 9. If a slot being blocked already has a confirmed booking: the system alerts the Veterinarian and requires them to manually cancel or reschedule the affected appointment before the block takes effect
 10. The Veterinarian may view the full medical history and previous visit records of a client's animal before or during an appointment, including past prescriptions, diagnoses, and uploaded files

- Workflow / Sequence of upcoming Events
1. The Veterinarian navigates to their personalized dashboard displaying appointments sorted chronologically
 2. The system displays appointments for the day, week, or month with the service type shown as a label on each entry
 3. The Veterinarian selects an appointment and updates its status through the defined stages: Confirmed, En Route, In Progress, or Completed
 4. The system automatically notifies the Client via the platform whenever the appointment status changes
 5. Upon marking a visit as Completed: the system prompts the Veterinarian to upload visit notes (free text, max 2,000 characters), a digital prescription (PDF or structured form), and any relevant files (photos, lab results; max 10 MB per file)
 6. The uploaded records are securely stored and attached to the client's animal profile and made accessible to the Client through their account
 7. If the Veterinarian wishes to flag the appointment for follow-up: they specify a follow-up date and the system automatically sends a reminder notification to both the Veterinarian and the Client on that date
 8. The Veterinarian may also set or update their availability calendar by marking specific days or time slots as unavailable
 9. If a slot being blocked already has a confirmed booking: the system alerts the Veterinarian and requires them to manually cancel or

	reschedule the affected appointment before the block takes effect 10. The Veterinarian may view the full medical history and previous visit records of a client's animal before or during an appointment, including past prescriptions, diagnoses, and uploaded files
Post-conditions / Response	1. The appointment status is updated and the Client is notified of the change 2. Post-visit medical records and prescriptions are securely stored and linked to the client's animal profile 3. Follow-up reminders are scheduled and will be sent automatically on the specified date 4. Availability calendar changes are reflected in real time in the booking system
Comments	Access to animal medical history is restricted to records associated with animals linked to the Veterinarian's own appointments only. Visit notes are limited to 2,000 characters and file attachments are limited to 10 MB per file

5 Use Case: Submit Review

Individual Task — Fadi Bassous

System Name : Vet Doctor – Veterinary Home Visit Service	
Use Case Title	SUBMIT REVIEW
Brief Description	A registered Client may submit feedback for a completed veterinary visit by providing a rating and a written review. The system verifies that the visit has been completed, records the review, updates the veterinarian's rating, and makes the review visible according to the system's review policy.
Actors	Client
Data	Appointment details, Client information, Veterinarian information, rating value, written review, review date, review status
Stimulus / Trigger	User command issued by the Client upon selecting “ <i>Submit Review</i> ” for a completed appointment
Pre-conditions	1. The Client is registered and logged into the system 2. The Client has a completed appointment in the system 3. The Client has not already submitted a review for the same appointment

Workflow / Sequence of upcoming Events	1. The Client navigates to the completed appointment 2. The Client selects “<i>Submit Review</i>” 3. The system displays the review form 4. The Client enters a rating value 5. The Client may enter a written review 6. The Client submits the review 7. The system validates that the appointment is completed 8. The system checks that the Client has not already submitted a review for the same appointment 9. The system stores the review and links it to the appointment and Veterinarian 10. The system updates the Veterinarian’s average rating 11. The system displays a confirmation message to the Client
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Post-conditions / Response	1. The review is recorded in the system 2. The review is linked to the completed appointment and assigned Veterinarian 3. The Veterinarian’s average rating is updated 4. The Client cannot submit another review for the same appointment
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Comments	Only registered Clients may submit reviews. Reviews can only be submitted for completed appointments. Each completed appointment may receive only one review from the Client.
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